BACHELOR THESIS

Olivier Amacker

⚠️ ATTENTION REQUIRED — review before submission

— The thesis reference (`thesis-id`) has not been updated.
— The signature image (`signature`) is missing.
— The title is too long and wraps onto too many lines.

Numerical Stability of Functional MRI Connectivity Biomarkers

Informatique et Systèmes de communication (ISC)

Thesis reference ISC-ID-26-1

University of Applied Sciences and Arts Western Switzerland
Faculty of Engineering and Architecture
Haute École d'Ingénierie de Sion

Informatique et systèmes de communication ISC

Olivier Amacker

Numerical Stability of Functional MRI Connectivity Biomarkers

Thesis submitted to the Bachelor degree programme
Informatique et Systèmes de communication (ISC) – Data engineering major
Haute École d'Ingénierie de Sion

Supervisor: **PROF. DR OSCAR ESTEBAN**
Co-supervisor: **DR OKITO YAMASHITA**

Expert: **DR AYUMU YAMASHITA**

Submitted on August 11, 2026. Revision 0.0.1

PROJECT REPOSITORY
> Go to **github**

Table of Contents

Abstract

TODO Remove

- The context and motivation for the research.
- The main objective or research question.
- A brief description of the methodology or approach used.
- The key results or findings.
- The main conclusion or implications of the work.

The abstract should be self-contained, clear, and usually does not exceed 250–300 words. It allows readers to quickly understand the purpose and outcomes of the thesis without reading the full document.

The abstract **must** be written in both French and English.

TODO Double check

As Functional Magnetic Resonance Imaging (fMRI) pipelines become more computationally heavy, requiring high-dimensional computations, it is crucial to assess how small numerical variability introduced by several different factors like the running Operating System (OS), parallelization strategies, and hardware architecture affects the results of these pipelines.

Building upon the paper "Numerical Variability of Magnetic Resonance Imaging (MRI) Graph Measures" [1], this work goes a step further than looking at different MRI graph measures by also looking at Major Depressive Disorder (MDD) biomarkers, specifically the Principal Component Analysis (PCA) based biomarkers proposed by the paper "Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection" [2] by perturbing several steps of the pipeline. It also assesses the impact of numerically perturbated fMRIPrep preprocessing on FC Functional Connectivity (FC) matrices using the Numerical Population Variability Ratio (NPVR) metric.

To reproduce the results of the original study [1], the results were obtained by running the fMRIPrep [3] (25.2.5) preprocessing pipeline through a custom Docker container with floating-point arithmetic perturbations introduced by the Verificarlo [4] and Fuzzy [5] libraries. After preprocessing, FC matrices were obtained using Nilearn [6] (0.13.1). Different absolute thresholds were applied to the FC matrices, and graph metrics were computed using NetworkX [7] (3.6.1).

For the assessment of the PCA feature selection, two steps of the pipeline were perturbated: the correlation coefficient computation using a specific Fuzzy Docker image, and the principal component analysis by forcing it to use 32-bit floating-point inputs rather than 64-bit.

The results given by the original work reproduction closely matched the original ones, except for a specific graph where the Y-axis values were hardcoded inside the original code. The PCA feature extraction appeared to be very stable under the introduced numerical perturbations, simulating OS level numerical variability. Giving the exact same results.

These findings suggest that the PCA features extraction is not only stable across different imaging sites and datasets, but also robust to numerical variability.

Keywords
medical data science, fMRI, fMRIPrep, numerical stability, bio-markers, PCA, neuroimaging, MDD

DÉPÔT DU PROJET
> Lien **github**

Résumé

TODO Double check

Alors que les pipelines de Imagerie par Résonance Magnétique Fonctionnelle (IRMf) deviennent de plus en plus lourdes computationnellement, nécessitant des calculs en haute dimension, il est crucial d'évaluer comment de petites variations numériques introduites par plusieurs facteurs tels que le Système d'Exploitation (SE) utilisé, les stratégies de parallélisation et l'architecture matérielle affectent les résultats de ces dernières.

S'appuyant sur le papier « Numerical Variability of IRMf Graph Measures » [1], ce travail va plus loin que l'examen de différentes mesures de graphes d'Imagerie par Résonance Magnétique (IRM) en examinant également les biomarqueurs du Trouble Dépressif Majeur (TDM), notamment les biomarqueurs basés sur l'Analyse en Composantes Principales (ACP) proposés par le papier "Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection" [2], en perturbant plusieurs étapes du pipeline. Il évalue également l'impact du prétraitement fMRIPrep numériquement perturbé sur les matrices de Connectivité Fonctionnelle (CF) en utilisant la métrique Numerical Population Variability Ratio (NPVR).

Pour reproduire les résultats de l'étude originale [1], les résultats ont été obtenus en exécutant le pipeline de prétraitement fMRIPrep [3] (25.2.5) dans un conteneur Docker personnalisé avec des perturbations arithmétiques en virgule flottante introduites par les bibliothèques Verificarlo [4] et Fuzzy [5]. Après prétraitement, les matrices de CF ont été obtenues à l'aide de Nilearn [6] (0.13.1). Différents seuils absolus ont été appliqués aux matrices de CF, et les métriques de graphes ont été calculées à l'aide de NetworkX [7] (3.6.1).

Pour l'évaluation de la sélection de caractéristiques par ACP, deux étapes du pipeline ont été perturbées : le calcul des coefficients de corrélation à l'aide d'une image Docker Fuzzy spécifique, et l'analyse en composantes principales en la forçant à utiliser des entrées en virgule flottante 32 bits plutôt que 64 bits.

Les résultats de la reproduction de l'étude originale correspondent étroitement aux résultats originaux, à l'exception d'un graphe spécifique dont les valeurs de l'axe Y étaient codées en dur dans le code original. L'extraction de caractéristiques par ACP s'est avérée très stable sous les perturbations numériques introduites, simulant la variabilité numérique au niveau du SE, tout en produisant des résultats strictement identiques.

Ces résultats suggèrent que l'extraction de caractéristiques par ACP est non seulement stable entre différents sites d'imagerie et jeux de données, mais également robuste face à la variabilité numérique.

Mots-clés
medical data science, fMRI, fMRIPrep, numerical stability, bio-markers, PCA, neuroimaging, MDD

Acknowledgements

I want to thank the following people and institution which played a huge role in making this bachelor thesis possible:

- Prof. Dr Oscar Esteban, for giving me the opportunity to conduct this Bachelor Thesis abroad, as well as for his guidance and supervision.
- Dr Okito Yamashita and Dr Ayumu Yamashita, for their dedication and continuous feedback throughout this project.
- Dr Yohan Chatelain for his help on the Fuzzy perturbations
- The Advanced Telecommunications Research Institute International (ATR), for providing the necessary resources and environment to carry out this work.
- My family and friends back home, for their emotional support and encouragement during this journey.

TODO Remove

Writing a report is an exercise that involves both **content** and **form**. In this document, we aim to simplify the formatting aspect without making any assumptions about the content, specifically in the context of the ISC degree programme[1].

0.1 – The content of a thesis

The general structure of a bachelor thesis typically includes the following sections:

1. **Abstract**: A concise summary of the thesis, including the research question, methodology, results, and conclusions.
2. **Résumé**: A summary of the thesis in French.
3. **Acknowledgements**: [Optional] A section to thank those who supported your work.
4. **Table of Contents**: An organized list of chapters and sections.
5. **Introduction**: Presents the background/context, motivation, objectives, and scope and plan of the thesis.
6. **State of the Art / Literature Review**: Reviews existing research and situates the thesis within the academic context, if relevant to your work.
7. **Development and Methodology**: Describes the methods, materials, and procedures used in the research/thesis.
8. **Results**: Presents the findings of the research, often with tables, figures, and analysis.
9. **Discussion**: Interprets the results, discusses implications, and relates findings to the research question.
10. **Conclusion**: Summarizes the main findings and contributions, and suggests future work.
11. **References / Bibliography**: Lists all sources cited in the thesis.
12. **Appendices**: (Optional) Contains supplementary material such as raw data, code, or additional explanations.

[1] Here is how to add a footnote https://isc.hevs.ch

This structure may vary depending on the field of study, but these elements are commonly found in most bachelor theses. They are recommended for the *ISC Bachelor thesis* and should be adapted to the specific requirements of your thesis (e.g., if you have a state of the art section or not).

You can also change the order or the names of the sections, for instance, if you want to put the state of the art before the introduction, or if you want to add a section on methodology before the results.

0.2 – Academic titles

Please note that the academic titles of your supervisors and experts are important.

They should be included on the cover page, and you should use the correct title when addressing them in the acknowledgements section. For instance, a professor should be addressed as “Prof. [Name]”, while a doctor should be addressed as “Dr [Name]” (**without a colon!**). A professor who is also a doctor should be addressed as “Prof. Dr [Name]”.

If you are unsure about the title of your supervisor, co-supervisor, or expert, you can ask them directly or check their profile on the university website.

0.3 – Compiling the thesis

If you compile your thesis using the `typst` command line tool, or by using the `typst` extension in Visual Studio Code, please note that you must install the fonts used in this template. You can do so by running the following command in your terminal:

```
./fonts/install_fonts.sh
```

Chapter 1

Introduction

Functional Magnetic Resonance Imaging (fMRI) has revolutionized neuroscience by enabling non-invasive observation of brain activity through blood oxygenation level-dependent signals. A key application is the analysis of Functional Connectivity (FC), which examines temporal correlations between spatially distinct brain regions to construct connectivity matrices. These matrices serve as the foundation for deriving graph-theoretical metrics and biomarkers used in clinical research, particularly for conditions such as Major Depressive Disorder (MDD).

As fMRI pipelines become increasingly computationally intensive, involving high-dimensional matrix operations and complex preprocessing steps, the question of numerical reliability becomes crucial. Small numerical perturbations introduced by factors such as Operating System (OS) differences (as you can see on Figure 1), hardware architecture, and parallelization strategies can propagate through the pipeline, potentially affecting downstream results.

Figure 1 - Example of the same program giving different results depending on the OS

In the context of multi-site studies and clinical applications, where reproducibility is essential, understanding and quantifying this numerical variability is critical. Unstable biomarkers could lead to unreliable clinical decisions or failed replication across research sites. However, systematic assessment of numerical stability in fMRI pipelines remains limited, particularly for advanced feature extraction methods.

The goal of this bachelor thesis, conducted in collaboration with the Advanced Telecommunications Research Institute International (ATR), is to investigate the numerical stability of fMRI connectivity biomarkers. This work was divided into three complementary parts.

First, the objective was to reproduce the results of Alizadeh et al., 2026 [1], which examined how numerical variability affects the sample size required for statistical significance and the stability of FC matrix graph metrics using the Numerical Population Variability Ratio (NPVR) metric.

Second, the focus shifted to the numerical stability of the FC matrices themselves, rather than their derived graph metrics, providing edge-wise stability analysis.

Finally, the numerical stability of a feature extraction method based on Principal Component Analysis (PCA) was assessed, building on the work of Yamashita et al., 2026 [2]. This method extracts FC biomarkers robust to different sites and datasets. The stability assessment involved perturbing the correlation coefficient computation (`np.corrcoef`) and forcing the PCA to use 32-bit floating-point inputs rather than 64-bit.

Chapter 2

State of the Art

In recent years, different tools have been created to help the assessment of numerical variability. Notably Verificarlo Denis et al., 2018 [4] which is a tool used to compile programs with LLVM level arithmetic perturbations and Fuzzy [5] which consists of already perturbed Docker containers for certain Python libraries like Numpy or Scipy, and more recently Fuzzy PyTorch Gonzalez-Pepe et al., 2026 [8] which extends this approach to deep learning workflows.

These tools rely on Monte Carlo Arithmetic (MCA) Parker et al., 1997 [9], a technique that introduces controlled random perturbations to floating-point operations during program execution. By repeatedly running the same pipeline with MCA instrumentation, researchers can quantify how numerical variability propagates through computational workflows.

Using these tools, the impact of numerical variability has been investigated across several neuroimaging domains such as functional, structural and diffusion imaging. As introduced in Section 1, Alizadeh et al., 2026 [1] evaluated the numerical variability of different graph measures derived from the FC matrices resulting from the widely used fMRIPrep Esteban et al., 2019 [3] preprocessing pipeline. Using the NPVR metric, they obtained values ranging from 0.1 to 0.2 for most graph metrics. These results were found to vary across brain regions, FC thresholds and confound regression strategies.

In structural imaging pipelines, Mirhakimi et al., 2025 [10] investigated numerical uncertainty in linear registration algorithms, demonstrating that small floating-point variations can lead to measurable differences in image alignment. Similarly, Chatelain et al., 2026 [11] quantified the impact of numerical variability on structural Magnetic Resonance Imaging (MRI) measures in Parkinson's disease, finding that numerical variation reached nearly one-third of population variability in multiple cortical and subcortical regions.

In diffusion imaging pipelines, Kiar et al., 2021 [12] showed that numerical uncertainty in analytical workflows leads to impactful variability in brain network measurements, revealing that different computational environments can produce substantially different connectivity results.

Interestingly, numerical variability has also been used as a resource rather than a limitation. Gonzalez-Pepe et al., 2026 [8] established that controlled numerical perturbations can serve as an effective data augmentation strategy for deep learning models, artificially expanding training datasets and improving model generalization.

However, these stability conclusions cannot be generalized across pipelines due to heterogeneous methodologies and implementations. While graph-theoretical metrics have been studied by Alizadeh et al., 2026 [1], edge-wise FC matrix stability remains unexplored. Additionally, the PCA-based feature extraction proposed by Yamashita et al., 2026 [2] has shown robustness across different sites and datasets, making its numerical stability verification crucial. This thesis addresses these gaps through edge-wise FC analysis and PCA-based extraction evaluation.

Chapter 3

Development and Methodology

This section describes the development process and methodology underlying this thesis. Beyond the thesis-specific code, this work included contributions to external repositories via Pull Requests (PRs), including fixes to the ISC-HEI Typst thesis template, enhancements to Yamashita et al.'s PCA-based feature extraction package, and a correction to the NPVR simulation in Alizadeh et al.'s GitHub repository.

3.1 – Code availability

The code developed during this bachelor thesis is publicly available on GitHub. The repository is organized using git submodules to separate concerns: the `bachelor-thesis-report` submodule contains this Typst report, while the `bachelor-thesis-project` submodule contains all analysis code, including preprocessing pipelines, analysis notebooks, and job submission scripts for the ATR High Performance Computing (HPC) cluster.

An interactive project website is available at olivier.amacker.dev/bachelor-thesis, providing browsable access to the four analysis notebooks developed with Marimo and their respective outputs, a daily journal documenting the project's progress, and the different presentations delivered during the thesis.

The development environment is managed via Nix and uv, ensuring its reproducibility. The fuzzy fMRIPrep Docker container, built on Verificarlo and the Fuzzy's libmath library, used for running perturbed preprocessing runs, is published on DockerHub for reproducibility.

3.2 – Reproduction of Alizadeh et al., 2026

This section details the reproduction of the two distinct analyses presented in Alizadeh et al., 2026 [1]. The original code can be found on GitHub, and before starting the reproduction, a comprehensive summary of the paper was created to ensure a thorough understanding of the methodology, preprocessing pipeline, and graph metrics which is accessible here. The first analysis involved implementing NPVR calculation on simulated data to assess how NPVR variation influences the sample size required for Cohen's d coefficient. The second analysis used perturbed fMRIPrep outputs to evaluate NPVR variation across different graph metrics, thresholds, and brain regions.

3.2.1 – NPVR Simulation

An interactive notebook for this simulation can be accessed online, as it doesn't rely on any external data.

The simulation begins by generating two synthetic populations using normal distributions with the same means but different standard deviations: $\sigma = 0.1$ for the low variability group and $\sigma = 0.4$ for the high variability group. For each population, the NPVR is calculated using the formula $\mathrm{NPVR} = \frac{\sigma_{\mathrm{num}}}{\sigma_{\mathrm{pop}}}$, where σ_{num} represents the numerical variability and σ_{pop} the population variability. The populations are then visualized in the σ_{num} / σ_{pop} space to assess their relative positions on NPVR contour lines. Finally, the relationship between NPVR, sample size, and Cohen's d variations is visualized to assess how numerical noise propagates into statistical inference.

During this reproduction, the original source code was reviewed and an error was identified. The NPVR values in the final visualization were hardcoded to `0.287` and `0.496` rather than computed from the generated distributions. This bug was fixed through a PR that was merged into the original repository.

3.2.2 – Graph Metrics Stability Assessment

The notebook code developed for this step and its outputs are available as a static HTML page.

This reproduction step involved several changes to the original procedure. First, a custom Docker container was created to perturb the latest version of fMRIPrep (25.2.5) instead of using the existing Fuzzy container, which used an older fMRIPrep version (23.2.1). Second, the fMRI data used for this section was provided by the ATR and originates from a multi-site dataset Koike et al., 2021 [13], rather than the publicly available PPMI dataset that was originally used. Third, when extracting the FC matrices, the original paper used six confound regressors corresponding to the main motion parameters (translations and rotations). However, as the data used in this work appeared to be noisier, nine additional confounds were included following the recommendation of Dr. Yamashita: global signal, CSF, white matter, and six aCompCor components (`a_comp_cor_00` through `a_comp_cor_05`). Fourth, due to computational constraints, small-worldness was excluded from the analysis. Finally, how the NPVR is calculated based on the outputs of the different fMRIPrep runs was modified. In the original paper, they had the same number of runs per subject and the NPVR was calculated as follows:

$$\sigma^2_{\text{num}} = \frac{1}{m}\sum_{j=1}^{m}\left[\frac{1}{n-1}\sum_{i=1}^{n}\left(x_{i,j} - x - |_{.,\text{j}}\right)^2\right]$$

$$\sigma^2_{\text{pop}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{1}{m-1}\sum_{j=1}^{m}\left(x_{i,j} - x - |_{i,.}\right)^2\right]$$

$$\text{NPVR} = \frac{\sigma_{\text{num}}}{\sigma_{\text{pop}}}$$

where n is the number of MCA repetitions, m is the number of subjects, and $x_{i,j}$ is the metric value for subject j in MCA run i.

Initially, some preprocessing runs failed, resulting in varying numbers of MCA runs per subject. However, after rerunning the failed subjects, the final dataset contained 5 subjects with 5 fuzzy fMRIPrep runs each. Although the final dataset was balanced, the pooled approach was retained to handle potential missing data robustly. The pooled numerical variance accounts for the varying number of runs per subject:

$$\sigma^2_{\text{num pooled}}(r) = \frac{1}{\sum_{j=1}^{m}(n_j - 1)}\sum_{j=1}^{m}(n_j - 1)\left[\frac{1}{n_j - 1}\sum_{i\in\Omega_j}\left(x_{i,j}(r) - x - |_{.,\text{j}}(r)\right)^2\right]$$

Similarly, the pooled population variance accounts for the varying number of subjects per run:

$$\sigma^2_{\text{pop pooled}}(r) = \frac{1}{\sum_{i=1}^{n}(m_i - 1)}\sum_{i=1}^{n}(m_i - 1)\left[\frac{1}{m_i - 1}\sum_{j\in\Omega_i}\left(x_{i,j}(r) - x - |_{i,.}(r)\right)^2\right]$$

The pooled NPVR for each region r is then:

$$\text{NPVR}_{\text{pooled}(r)} = \frac{\sigma_{\text{num pooled}(r)}}{\sigma_{\text{pop pooled}(r)}}$$

where n_j is the number of runs for subject j, m_i is the number of subjects in MCA run i, and Ω_j and Ω_i denote the sets of valid indices for subject j and run i, respectively.

This approach ensures that all available data contributes to the variability estimates, even if some runs are missing.

The remaining analysis followed the same procedure as the original study. After fMRIPrep preprocessing, the Schaefer 2018 parcellation [14] with 100 cortical regions and 7 functional networks was used to extract regional time series using Nilearn's NiftiLabelsMasker. Spatial smoothing of 6mm Full Width at Half Maximum (FWHM), temporal standardization, and detrending were applied during masking. The FC matrices were computed as Pearson correlation matrices using Nilearn's ConnectivityMeasure, with both with-confound and without-confound versions generated. The matrices were thresholded using absolute correlation values at six thresholds: 0.05, 0.1, 0.2, 0.3, 0.4, and 0.5, retaining both strongly positive and strongly negative correlations. For each threshold, four local graph metrics were computed using NetworkX: degree centrality, clustering coefficient, betweenness centrality, and eigenvector centrality. Additionally, one global metric was calculated: average shortest path length. The NPVR was then computed for each metric and threshold, and plotted across brain regions by projecting regional values onto the Schaefer 2018 atlas to assess spatial variability in numerical stability. Additionally, the NPVR was computed on the difference between with-confound and without-confound matrices to assess the impact of confound regression, following the approach of the original study.

3.3 – Edge-wise FC Matrix Stability Analysis

The notebook code developed for this step and its outputs are available as a static HTML page.

Reusing the perturbed fMRIPrep runs from Section 3.2.2, this section extends the analysis from graph-level metrics to individual edges of the FC matrices. The pooled NPVR was computed independently for each of the 4950 edges (upper triangle of the 100x100 matrix) using the same formula as before.

Two confound strategies were compared: the full set of 15 confounds and a filtered set excluding only the global signal. The global signal regression remains a debated practice in the neuroimaging community, as it may remove both nuisance signals and neural activity of interest Xu et al., 2018 [15], Xifra-Porxas et al., 2025 [16]. To quantify its impact on edge-wise numerical stability, a delta NPVR was computed for each edge as the difference between the two confound strategies.

The edge-wise NPVR values were visualized as connectivity matrices, scatter plots, and histograms. To assess spatial patterns, edge-wise values were aggregated per brain region by computing the mean and median NPVR across all connected edges and mapped onto the Schaefer 2018 atlas.

3.4 – PCA-based Feature Extraction Stability

A browsable version of the analysis notebook is available online.

This section had three objectives. First, reproduce the PCA-based feature extraction method developed by Yamashita et al., 2026 [2] on the complete SRPB dataset [17], as opposed to the original paper which split the data into discovery and validation sets. Second, apply the same method to the complete BMB dataset [13]. Third, perturb different steps of the pipeline: the correlation computation using a MCA-instrumented Docker container and the PCA dimensionality reduction by forcing 32-bit floating-point inputs instead of the standard 64-bit precision.

3.4.1 – Reproduction on the SRPB and BMB Datasets

The original PCA feature extraction code was publicly available on GitHub but organized as separate Python files, making integration challenging. A PR was submitted and merged to restructure it into the `pcafeat` package with uv as a dependency manager.

The FC matrices used as inputs to the pipeline had already been preprocessed and harmonized across imaging sites by ATR for both datasets using the method of Yamashita et al., 2019 [18], so they were used directly without any additional preprocessing or harmonization step.

The extraction pipeline was applied to six metrics: diagnosis (MDD vs. HC), BDI score, age, sex, imaging site, and mean Framewise Displacement (FD). The pipeline works as follows:

1. A PCA reduces all n subjects' FC vectors to n Principal Components (PCs)
2. Each PC is tested for association with the desired target variables using appropriate statistical tests, with results plotted during testing:
   - t-tests for binary categories (e.g., sex)
   - ANOVA for multi-level categories (e.g., site)
   - Pearson's correlation for continuous variables (e.g., age)
3. PCs significantly associated with the target after False Discovery Rate with Benjamini-Hochberg (FDR-BH) correction $(q < 0.05)$ are selected
4. Their FC weights are tested via a chi-squared test with the same correction threshold
5. Only FCs surviving the corrections are retained as final biomarkers

During replication of the original paper's pipeline, the plots generated during PC association testing were found to have missing labels and incomplete titles. This issue was fixed in another PR.

After running the extraction pipeline on both datasets, the selected FCs for the second PC were plotted on brain surface maps using a combined parcellation of 446 Regions Of Interest (ROIs): 360 cortical regions from the Glasser et al., 2016 [19] atlas, 54 subcortical regions from the Tian et al., 2020 [20] atlas, and 32 cerebellar regions from the Nettekoven et al., 2024 [21] atlas. Functional network assignments were derived from the Yeo et al., 2011 [22] 7-network parcellation. Each selected FC connection was visualized as an edge between brain regions, color-coded by direction of effect: red indicating over-connectivity in MDD relative to healthy controls, and blue indicating under-connectivity. Nodes were colored according to their functional network assignment (Visual, SomatoMotor, DorsalAttention, DefaultMode, Limbic, Salience, PrefrontalControl, Subcortical, or Cerebellum).

Network statistics were also computed to quantify the distribution of selected connections across brain networks, including the number of participating nodes per network, the proportion of intra-network versus inter-network edges, and the most frequent inter-network connection pairs.

3.4.2 – FC Matrix Regular Extraction

While Section 3.4.1 used FC matrices already preprocessed and harmonized by ATR, the perturbation strategy described in the next section targets the correlation coefficient computation (`np.corrcoef`) used during the FC matrices extraction. Reproducing the extraction and harmonization pipeline from the raw parcellated time series was therefore necessary to establish a numerically consistent pipeline before introducing any perturbation.

The starting point was the preprocessed regional time series provided by ATR for the SRPB dataset, parcellated according to the Glasser et al., 2016 [19] atlas (360 cortical, 54 subcortical, and 32 cerebellar regions, 446 ROIs in total) with global signal regression and band-pass filtering already applied.

FC extraction proceeded as follows. First, the regional time series were scrubbed for motion by removing any frame with FD exceeding 0.5 mm, together with the immediately subsequent frame. On the retained frames, Pearson correlation matrices were computed using `np.corrcoef` applied to the transposed time series array. The lower triangular elements of each correlation matrix were extracted, Fisher Z-transformed via `np.arctanh`, and flattened into a subject-level connectivity vector.

Before harmonization, the self-extracted FC matrices were validated against the original ATR-provided harmonized matrices. For each subject, the Pearson correlation and mean absolute error between the extracted and reference connectivity vectors were computed. The resulting per-subject correlation distribution showed near-perfect agreement, confirming that the extraction pipeline faithfully reproduced the original connectivity values prior to harmonization (Figure 2).

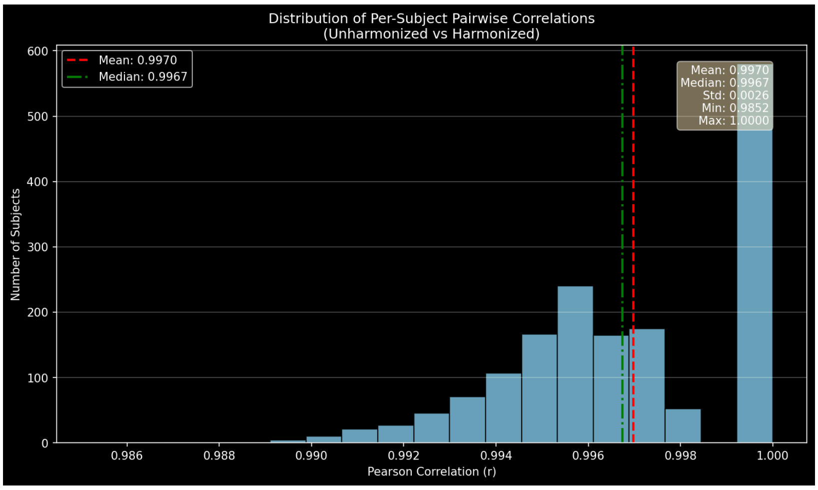

Figure 2 - Per-subject Pearson correlation distribution between self-extracted and ATR-provided FC connectivity vectors before harmonization.

Cross-site harmonization was then performed using the traveling-subject method of Yamashita et al., 2019 [18]. This approach models unwanted variance as a linear combination of subject, site, sampling, and protocol biases. Dummy variables were created for each categorical factor, and the sampling dummies were orthogonalized against the site dummies using a weighted projection to ensure independence between site and sampling effects. The bias estimation was formulated as a constrained regularized least-squares problem, solved efficiently via Cholesky decomposition of the Hessian and Schur-complement elimination of the Lagrange multipliers within the corresponding Karush-Kuhn-Tucker system, avoiding explicit matrix inversion. Generalized cross-validation was used to select the regularization hyperparameter λ.

Once the bias coefficients were estimated, the relevant site and protocol bias terms were subtracted from each regular subject's connectivity vector, yielding harmonized FC matrices. The self-harmonized matrices were again compared against the ATR-provided harmonized matrices. The near-zero mean difference and high per-subject correlation confirmed that the entire reproduction pipeline from scrubbed time series to harmonized connectivity was numerically consistent with the original ATR processing, thereby establishing a valid baseline for the subsequent perturbation experiments (Figure 3). The identical procedure was subsequently applied to the BMB dataset.

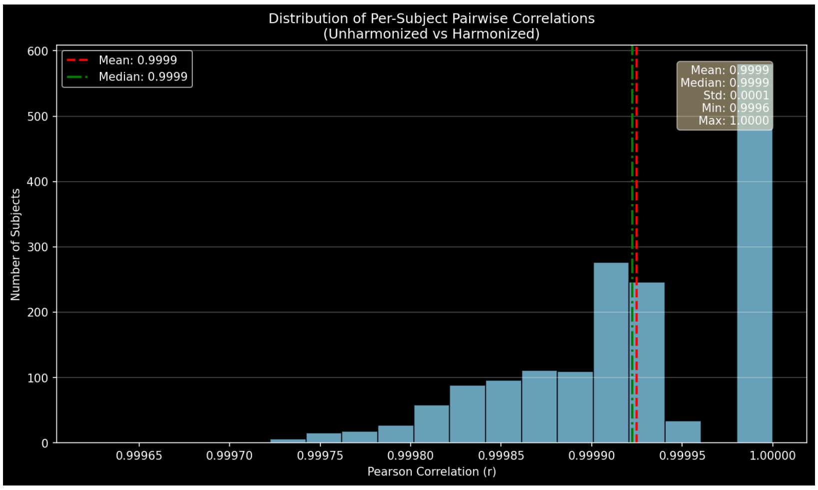

Figure 3 - Per-subject Pearson correlation distribution between self-harmonized and ATR-provided harmonized FC connectivity vectors.

3.4.3 – FC Matrix Perturbed Extraction

Once the regular FC extraction pipeline was validated, a perturbed variant was implemented by executing the `np.corrcoef` correlation computation inside a MCA-instrumented Docker container provided by Fuzzy. One hundred perturbed extractions were performed on the SRPB dataset ($n = 1667$), while fifteen perturbed extractions were performed on the provided subset of the BMB dataset ($n = 6371$). The lower number of runs for the BMB dataset reflects its larger subject count and corresponding computational cost. Since the individual extractions within each dataset are independent yet computationally demanding, they were executed in parallel, while the two datasets themselves were processed sequentially.

3.4.4 – PCA Dimensionality Reduction Perturbation

3.4.5 – Perturbed Results Comparison

Chapter 4

Results and Discussion

Chapter 5

Conclusion

Bibliography

[1] M. Alizadeh, Y. Chatelain, G. Kiar, and T. Glatard, “Numerical Variability of functional MRI Graph Measures,” *bioRxiv*, 2026, doi: 10.64898/2025.12.22.695524.

[2] A. Yamashita *et al.*, “Extraction of robust functional connectivity patterns across psychiatric disorders using principal component analysis-based feature selection,” *Imaging Neuroscience*, vol. 4, p. IMAG.a.1121, 2026, doi: 10.1162/IMAG.a.1121.

[3] O. Esteban *et al.*, “fMRIPrep: a robust preprocessing pipeline for functional MRI,” *Nature Methods*, vol. 16, p. , 2019, doi: 10.1038/s41592-018-0235-4.

[4] C. Denis, P. D. O. Castro, and E. Petit, “Verificarlo: checking floating point accuracy through Monte Carlo Arithmetic.” [Online]. Available: https://arxiv.org/abs/1509.01347

[5] G. Kiar *et al.*, “verificarlo/fuzzy: Fuzzy v3.5.1.” [Online]. Available: https://doi.org/10.5281/zenodo.20906259

[6] N. contributors, “nilearn.” [Online]. Available: https://github.com/nilearn/nilearn

[7] A. A. Hagberg, D. A. Schult, and P. J. Swart, “Exploring Network Structure, Dynamics, and Function using NetworkX,” in *Proceedings of the 7th Python in Science Conference*, G. Varoquaux, T. Vaught, and J. Millman, Eds., Pasadena, CA USA, 2008, pp. 11–15.

[8] I. Gonzalez-Pepe, H. Akhaddar, T. Glatard, and Y. Chatelain, “Fuzzy PyTorch: Rapid Numerical Variability Evaluation for Deep Learning Models.” [Online]. Available: https://arxiv.org/abs/2605.25991

[9] D. S. Parker and D. Langley, “Monte Carlo Arithmetic: exploiting randomness in floating-point arithmetic,” 1997. [Online]. Available: https://api.semanticscholar.org/CorpusID:2321215

[10] N. Mirhakimi, Y. Chatelain, T. Glatard, and J.-B. Poline, “Numerical Uncertainty in Linear Registration: An Experimental Study.” [Online]. Available: https://arxiv.org/abs/2508.00781

[11] Y. Chatelain, A. Soko-lowski, M. Sharp, J.-B. Poline, and T. Glatard, “The practical impact of numerical variability on structural MRI measures of Parkinson’s disease,” *bioRxiv*, 2026, doi: 10.64898/2026.01.09.698203.

[12] G. Kiar *et al.*, “Numerical uncertainty in analytical pipelines lead to impactful variability in brain networks,” *PLOS ONE*, vol. 16, p. e250755, 2021, doi: 10.1371/journal.pone.0250755.

[13] S. Koike *et al.*, “Brain/MINDS beyond human brain MRI project: A protocol for multi-level harmonization across brain disorders throughout the lifespan,” *NeuroImage: Clinical*, vol. 30, p. 102600, 2021, doi: https://doi.org/10.1016/j.nicl.2021.102600.

[14] A. Schaefer *et al.*, “Local-Global Parcellation of the Human Cerebral Cortex from Intrinsic Functional Connectivity MRI,” *Cerebral Cortex*, vol. 28, no. 9, pp. 3095–3114, 2018, doi: 10.1093/cercor/bhx179.

[15] H. Xu *et al.*, “Impact of global signal regression on characterizing dynamic functional connectivity and brain states,” *NeuroImage*, vol. 173, pp. 127–145, 2018, doi: https://doi.org/10.1016/j.neuroimage.2018.02.036.

[16] A. Xifra-Porxas, M. Kassinopoulos, P. Prokopiou, M.-H. Boudrias, and G. D. Mitsis, “Global signal regression reduces connectivity patterns related to physiological signals and does not alter EEG-derived connectivity,” *bioRxiv*, 2025, doi: 10.1101/2024.04.18.590163.

[17] S. C. Tanaka *et al.*, “A multi-site, multi-disorder resting-state magnetic resonance image database,” *Scientific Data*, vol. 8, no. 1, p. 227, 2021, doi: 10.1038/s41597-021-01004-8.

[18] A. Yamashita *et al.*, "Harmonization of resting-state functional MRI data across multiple imaging sites via the separation of site differences into sampling bias and measurement bias," *PLOS Biology*, vol. 17, no. 4, pp. 1–34, 2019, doi: 10.1371/journal.pbio.3000042.

[19] M. F. Glasser *et al.*, "A multi-modal parcellation of human cerebral cortex," *Nature*, vol. 536, no. 7615, pp. 171–178, 2016, doi: 10.1038/nature18933.

[20] Y. Tian, D. S. Margulies, M. Breakspear, and A. Zalesky, "Topographic organization of the human subcortex unveiled with functional connectivity gradients," *bioRxiv*, 2020, doi: 10.1101/2020.01.13.903542.

[21] C. Nettekoven *et al.*, "A hierarchical atlas of the human cerebellum for functional precision mapping," *Nature Communications*, vol. 15, p. , 2024, doi: 10.1038/s41467-024-52371-w.

[22] B. T. Thomas Yeo *et al.*, "The organization of the human cerebral cortex estimated by intrinsic functional connectivity," *Journal of Neurophysiology*, vol. 106, no. 3, pp. 1125–1165, 2011, doi: 10.1152/jn.00338.2011.

[23] Wellcome Centre for Human Neuroimaging, "Understanding MRI: What is functional MRI (fMRI)?." [Online]. Available: https://www.youtube.com/watch?v=4UOeBM5BwdY

[24] Zachary Cortex, "fMRI Explained (Functional Magnetic Resonance Imaging)." [Online]. Available: https://www.youtube.com/watch?v=IGwChiQJJHk

[25] La Science En Schémas, "IMAGERIE MÉDICALE #7 : L'IRM (Imagerie par résonance magnétique)." [Online]. Available: https://www.youtube.com/watch?v=q_H5GRC48KU

[26] Olivier Amacker, "Semester project." [Online]. Available: https://github.com/MadeInShineA/303.1-semester-project

[27] T. Glatard, "Fuzzy fMRIPrep." [Online]. Available: https://github.com/glatard/fuzzy-fmriprep

[28] Verificarlo, "Fuzzy." [Online]. Available: https://github.com/verificarlo/fuzzy

[29] Enthought, "Fuzzy environments for the perturbation evaluation and appl of num uncertainty via Mo | SciPy 2021." [Online]. Available: https://www.youtube.com/watch?v=3zgS8nYaKJ0

[30] Verificarlo, "Verificarlo." [Online]. Available: https://github.com/verificarlo/verificarlo

[31] N. Terzimehić, B. Bühler, and E. Kasneci, "Conversational AI as a catalyst for informal learning: An empirical large-scale study on LLM use in everyday learning," *Computers and Education: Artificial Intelligence*, p. 100634, 2026, doi: https://doi.org/10.1016/j.caeai.2026.100634.

[32] N. Kosmyna *et al.*, "Your Brain on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task." [Online]. Available: https://arxiv.org/abs/2506.08872

[33] K. Moon, A. E. Green, and K. Kushlev, "Homogenizing effect of large language models (LLMs) on creative diversity: An empirical comparison of human and ChatGPT writing," *Computers in Human Behavior: Artificial Humans*, vol. 6, p. 100207, 2025, doi: https://doi.org/10.1016/j.chbah.2025.100207.

[34] J. Becker, N. Rush, E. Barnes, and D. Rein, "Measuring the Impact of Early-2025 AI on Experienced Open-Source Developer Productivity." [Online]. Available: https://arxiv.org/abs/2507.09089

[35] A. Alansari and H. Luqman, "Large Language Models Hallucination: A Comprehensive Survey." [Online]. Available: https://arxiv.org/abs/2510.06265

Appendix

Table of Acronyms

ACP : Analyse en Composantes Principales
ATR : Advanced Telecommunications Research Institute International
CF : Connectivité Fonctionnelle
FC : Functional Connectivity
FD : Framewise Displacement
FDR-BH : False Discovery Rate with Benjamini-Hochberg
FWHM : Full Width at Half Maximum
HPC : High Performance Computing
IRM : Imagerie par Résonance Magnétique
IRMf : Imagerie par Résonance Magnétique Fonctionnelle
MCA : Monte Carlo Arithmetic
MDD : Major Depressive Disorder
MRI : Magnetic Resonance Imaging
NPVR : Numerical Population Variability Ratio
OS : Operating System
PC : Principal Component
PCA : Principal Component Analysis
PR : Pull Request
ROI : Region Of Interest
SE : Système d'Exploitation
TDM : Trouble Dépressif Majeur
fMRI : Functional Magnetic Resonance Imaging

List of Figures

Code Appendix

```scala
class Cons[+A](hd: A, tl: => MyStream[A]) extends MyStream[A] {
  override def isEmpty: Boolean = false
  override val head: A = hd
  override lazy val tail: MyStream[A] = tl
  override def #::[B >: A](elem: B): MyStream[B] =
    new Cons[B](elem, this)

  override def ++[B >: A](anotherStream: => MyStream[B]): MyStream[B] =
    new Cons[B](head, tail ++ anotherStream)

  override def foreach(f: A => Unit): Unit = {
    f(head)
    tail.foreach(f)
  }
}
```

Listing 1 - Code included from the file example.scala

```python
def mergeSort(arr):
  if len(arr) <= 1:
    return arr

  mid = len(arr) // 2
  leftHalf = arr[:mid]
  rightHalf = arr[mid:]

  sortedLeft = mergeSort(leftHalf)
  sortedRight = mergeSort(rightHalf)

  return merge(sortedLeft, sortedRight)

def merge(left, right):
  result = []
  i = j = 0

  while i < len(left) and j < len(right):
    if left[i] < right[j]:
      result.append(left[i])
      i += 1
    else:
      result.append(right[j])
      j += 1

  result.extend(left[i:])
  result.extend(right[j:])

  return result

mylist = [3, 7, 6, -10, 15, 23.5, 55, -13]
mysortedlist = mergeSort(mylist)
print("Sorted array:", mysortedlist)
```

Listing 2 - Second code included from the file example.scala

```scala
class Cons[+A](hd: A, tl: => MyStream[A]) extends MyStream[A] {
  override def isEmpty: Boolean = false
  override val head: A = hd
  override lazy val tail: MyStream[A] = tl
```

```scala
  override def #::[B >: A](elem: B): MyStream[B] =
    new Cons[B](elem, this)

  override def ++[B >: A](anotherStream: => MyStream[B]): MyStream[B] =
    new Cons[B](head, tail ++ anotherStream)

  override def foreach(f: A => Unit): Unit = {
    f(head)
    tail.foreach(f)
  }
}
```

Listing 3 - Second code included from the file example.scala

Typeset with Typst
ISC-HEI template by P.-A. Mudry · v0.8.0
Informatique et systèmes de communication — Sion, HES-SO Valais

No LaTeX was harmed in the making of this document.